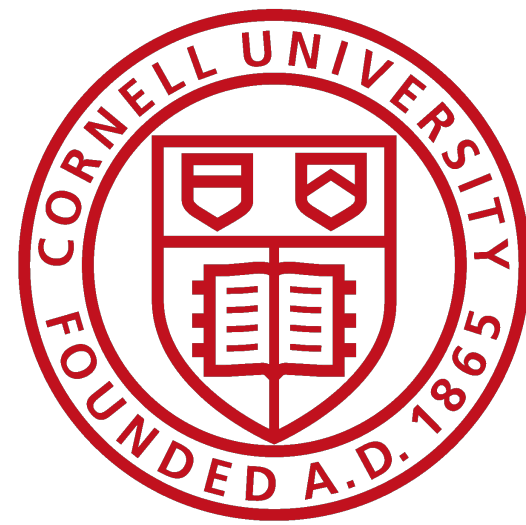




When Lines Become Contagious:

Queueing Theory for Disease Modelling

Laurel Newman, Jamol Pender

Stochastic Disease Models

Compartment models (e.g., SEIR model): individuals move between Susceptible, Exposed, Infectious, and Recovered states.

Population heterogeneity in disease modelling: trade-off between practical accuracy from simulations vs. theoretical rigor.

Population classes: chosen features modelled with different parameters as preliminary approach to heterogeneity.

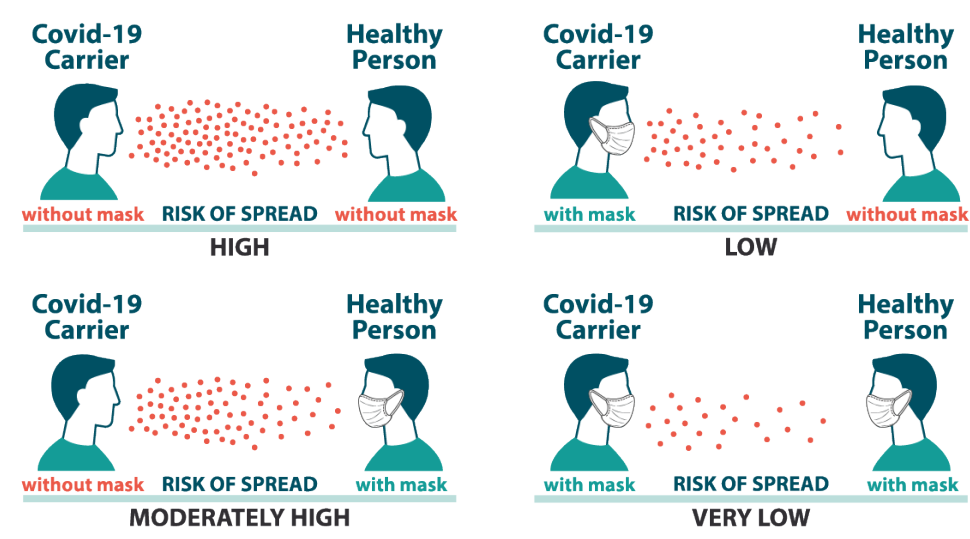
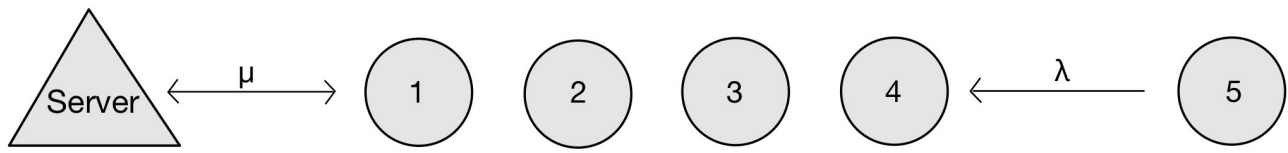


Figure 1. Two features each with two classes of customers gives us four interaction types.

Mathematical Background

$M/M/1$ queues: system with a single server, sequential customers, arrival times generated from a Poisson process with parameter λ , service times exponential with parameter μ .



Customer overlaps: in service systems, the entire population does not interact simultaneously.

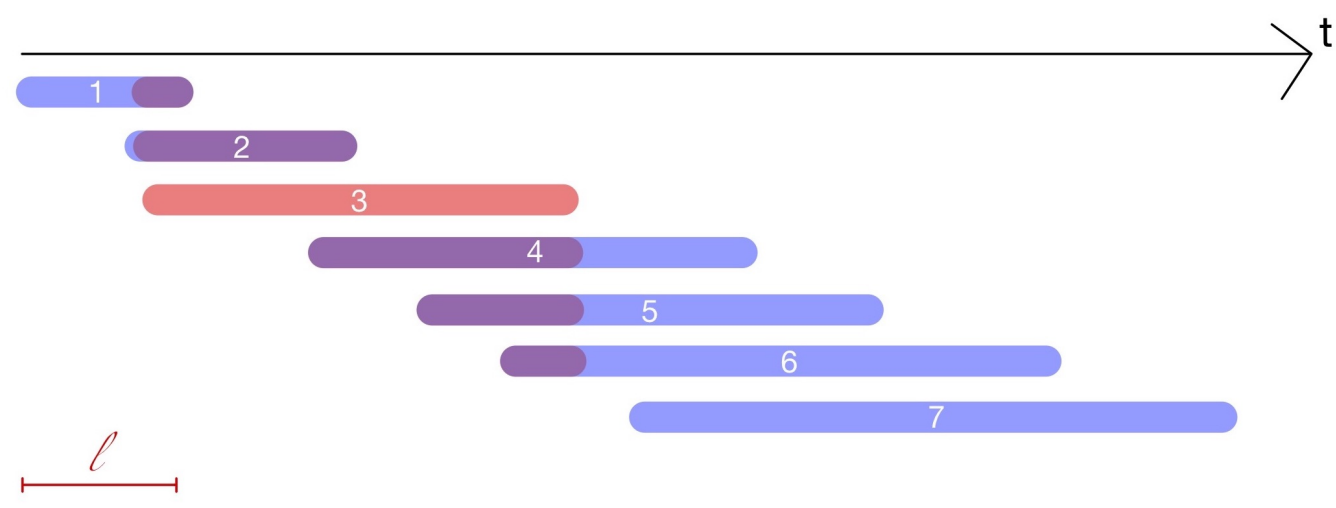


Figure 2. An overlap of at least ℓ with an infectious customer infects you!

Key Insight

Customers arrive infectious with probability p , and susceptible otherwise.

Dual perspectives: individual (people who infect us) and communal (people we infect).

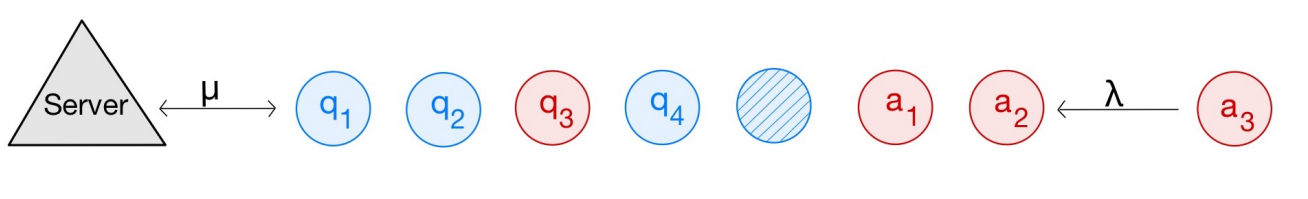


Figure 3. Susceptible customer perspective.

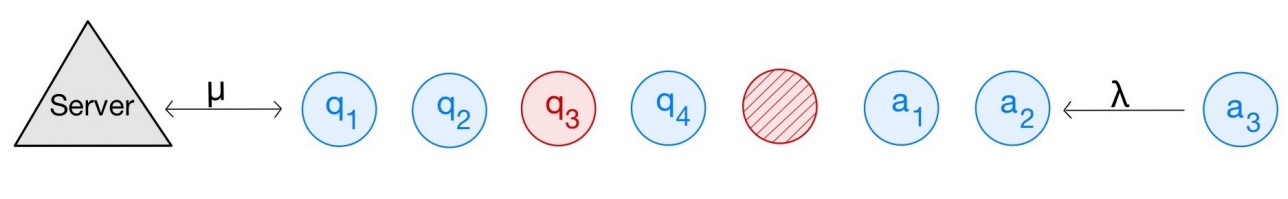


Figure 4. Infectious customer perspective.

Analyzing intervention efficacy: probability and severity of infection when we're unmasked vs. when we're masked.

Partitioning Approach

Separate problems: if we're masked or unmasked. Partition other customers: susceptible or infectious $\rightarrow$ masked or unmasked.

Partition into customers present in the queue upon our arrival, and the arrivals after us.

Modelling with Partitions

Our dual perspectives, see Figures 3 and 4, allow us to immediately simplify after arrivals.

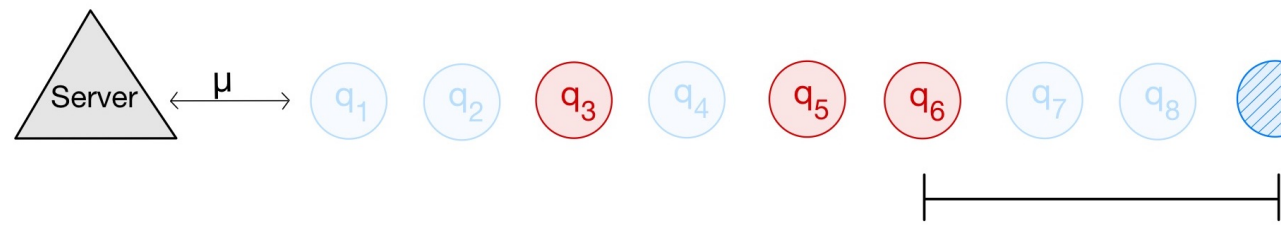


Figure 5. For a susceptible customer, upon arrivals reduces to considering closest infectious customer.



Figure 6. For infectious customers, upon arrivals reduces to first susceptible customer who overlaps at least ℓ .

Reduce to supposing we are susceptible and masking:

- p_u chance other customers are infectious and unmasked, with infection threshold ℓ_u ,
- p_m chance other customers are infectious and masking, with infection threshold ℓ_m ,
- $\ell_u < \ell_m$, and $p_u = p * (1 - q)$ while $p_m = p * q$, where q is masking probability.

Theorem 1: Susceptible customer infection probability

Let I be the event a customer gets infected while in the queue. Suppose a (masking) susceptible customer joins the $M/M/1$ queue in steady state, then the probability that the customer gets infected is equal to,

$$\mathbb{P}(I) = \frac{2\lambda p_u}{\mu - \lambda + 2\lambda p_u} e^{-(\mu - \lambda)\ell_u} \left(1 + \frac{p_m}{p_u} \frac{\mu - \lambda}{\mu - \lambda + 2\lambda p_u + 2\lambda p_m} e^{-(\mu - \lambda + 2\lambda p_u)(\ell_m - \ell_u)} \right).$$

This probability of infection is interpretable:

- First term multiplies two probabilities: sojourn time is at least unmasked infection threshold,
- and the exponential of our pass-through rate is more than the exponential of twice the unmasked infectious arrival rate.
- Then another product: system time without seeing unmasked infectious customer is at least masked infection threshold,
- and scaled similarly by exponential of twice the unmasked infectious arrival rate is more than the exponential of our pass-through rate, which is more than the exponential of twice the masking infectious arrival rate.

Twice the rate accounts for the equal probability of infection in both the upon and after arrival cases.

Theorem 2: Difference between exclusive infection probabilities

Let I_m be the event a customer only gets infected by a masked fellow customer while in the queue and I_u that they only get infected by an unmasked fellow customer:

$$\begin{aligned} \mathbb{P}(I_m) &= \frac{2\lambda p_m(\mu - \lambda)}{(\mu - \lambda + 2\lambda p_u)(\mu - \lambda + 2\lambda p_u + \lambda p_m)} e^{-(\mu - \lambda)\ell_m} e^{-\lambda p_u(\ell_m - \ell_u)} \\ &\quad \left(1 - \frac{\lambda p_m}{\mu - \lambda + 2\lambda p_u + 2\lambda p_m} e^{-\lambda p_u(\ell_m - \ell_u)} \right). \\ \mathbb{P}(I_u) &= \frac{2\lambda p_1}{\mu - \lambda + 2\lambda p_1} e^{-(\mu - \lambda)\ell_1} - \frac{2\lambda p_2}{\mu - \lambda + 2\lambda p_2} e^{-(\mu - \lambda)\ell_2} \\ &\quad \left(1 - \frac{(\mu - \lambda)(\mu - \lambda + 2\lambda p_2)}{(\mu - \lambda + 2\lambda p_1)(\mu - \lambda + 2\lambda p_1 + 2\lambda p_2)} e^{-2\lambda p_1(\ell_2 - \ell_1)} \right). \end{aligned}$$

Dual Perspective: Number Infected

Suppose an infectious customer joins the $M/M/1$ queue in steady state, then the m^{th} moment of the number of susceptible customers they infect is,

$$\mathbb{E}[N_I^m] = \frac{\mu - \lambda}{\mu - \lambda + 2\lambda(1 - p)} \text{Li}_{-m} \left(\frac{2\lambda(1 - p)}{\mu - \lambda + 2\lambda(1 - p)} \right),$$

where $\text{Li}_m(x)$ is the polylogarithm function and is defined by

$$\text{Li}_m(x) = \sum_{j=1}^{\infty} \frac{x^j}{j^m}.$$

Thus, the mean and variance of the number infected by one infectious customer is equal to

$$\begin{aligned} \mathbb{E}[N_I] &= \frac{2\lambda(1 - p)}{\mu - \lambda} e^{-(\mu - \lambda)\ell}, \text{ and} \\ \text{Var}[N_I] &= \frac{2\lambda(1 - p)(\mu - \lambda + 4\lambda(1 - p))}{(\mu - \lambda)^2} e^{-(\mu - \lambda)\ell} - \frac{4\lambda^2(1 - p)^2}{(\mu - \lambda)^2} e^{-2(\mu - \lambda)\ell}. \end{aligned}$$

Numerical Results

Let $q = 0.4$, $\ell_u = 0.1$, and $\ell_m = 0.4$. We average over 5 simulations, with 100,000 customers per each simulation.

λ	p	$\mathbb{P}(I)$ (form.)	$\mathbb{P}(I)$ (sim.)	$\mathbb{P}(I_u)$ (form.)	$\mathbb{P}(I_u)$ (sim.)	$\mathbb{P}(I_m)$ (form.)	$\mathbb{P}(I_m)$ (sim.)	λ	p	$\mathbb{P}(I)$ (form.)	$\mathbb{P}(I)$ (sim.)	$\mathbb{P}(I_u)$ (form.)	$\mathbb{P}(I_u)$ (sim.)	$\mathbb{P}(I_m)$ (form.)	$\mathbb{P}(I_m)$ (sim.)
.50	.01	.0176	.0169	.0111	.0107	.0063	.0060	.50	.10	.1498	.1500	.0891	.0885	.0483	.0486
.75	.01	.0536	.0544	.0324	.0342	.0197	.0184	.75	.10	.3551	.3576	.1802	.1816	.0986	.0970
.90	.01	.1492	.1465	.0847	.0853	.0528	.0496	.90	.10	.6291	.6351	.2269	.2278	.1171	.1115
.50	.02	.0346	.0337	.0217	.0217	.0123	.0114	.50	.50	.4502	.4503	.2163	.2162	.0983	.0941
.75	.02	.1015	.0100	.0600	.0585	.0361	.0363	.75	.50	.7117	.7103	.2181	.2164	.0917	.0845
.90	.02	.2590	.2587	.1380	.1384	.0834	.0837	.90	.50	.8813	.8827	.1294	.1267	.0507	.0445

Table 1. For a (masking) susceptible arrival in steady state, the theoretical and simulated probabilities that this customer becomes infected while in the system, and that they are specifically infected by just unmasked or masking other customers.

Extensions

$M/G/\infty$ Queues:

Infinite number of servers

General service times distribution

1. Susceptible customer infection probability
2. Higher moments of number infected

$M/G/\infty$ Comorbidity:

Multiple diseases in the system

1. Probability of contracting a subset
2. Probability of contracting only that subset
3. Which disease(s) contracted first/more times

Cybersecurity:

Side-channel attacks

1. After arrival probability
2. Overlapping sufficiently with service

In-System Infectiousness:

Become infectious after getting infected

Different Service Regimes:

Depending on prevalence of infection in system, prioritize serving infectious or susceptible people first.